

IB AAHL



FOR IB AAHL

HOW TO TEST CONTINUITY

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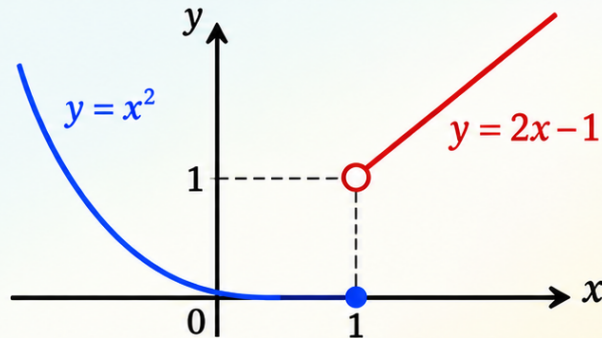
DIFFERENTIABILITY

IN

TWO STAGES

2

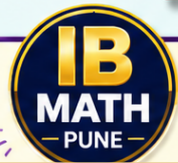
$$f(x) = \begin{cases} x^2, & x \leq 1 \\ 2x-1, & x > 1 \end{cases}$$



STAGE 1: TEST CONTINUITY



STAGE 2: TEST DIFFERENTIABILITY



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AHL 5.12

Content	Guidance, clarification and syllabus links
Informal understanding of continuity and differentiability of a function at a point.	In examinations, students will not be asked to test for continuity and differentiability.
Understanding of limits (convergence and divergence). Definition of derivative from first principles $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$	Link to: infinite geometric sequences (SL1.8). Use of this definition for polynomials only.
Higher derivatives.	Familiarity with the notations $\frac{d^n y}{dx^n}$, $f^{(n)}(x)$. Link to: proof by mathematical induction (AHL 1.15).

Stage (i): Continuity



A function f is **continuous at** $x = a$ if $f(a)$ and $\lim_{x \rightarrow a} f(x)$ exist and are equal.

f is also continuous at $x = a$ if:

- a is the left endpoint of an interval on which f is defined, and $f(a)$ and $\lim_{x \rightarrow a^+} f(x)$ exist and are equal, or
- a is the right endpoint of an interval on which f is defined, and $f(a)$ and $\lim_{x \rightarrow a^-} f(x)$ exist and are equal.

Question(1): Test for the continuity and differentiability of the following function at $x = 1$.

$$f(x) = \begin{cases} x^2, & x \leq 1 \\ 2x - 1, & x > 1 \end{cases} \quad \begin{matrix} \text{LHL} \\ \text{RHL} \end{matrix}$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x^2 = (1)^2 = 1$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (2x - 1) = 2(1) - 1 = 1$$

$$\text{Since } \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = 1$$

$\therefore f(x)$ is continuous at $x = 1$

Stage (ii):Differentiability



TESTING FOR DIFFERENTIABILITY

To test whether a function is differentiable, we first define:

- the **left-hand derivative** of f at $x = a$ is $f'_-(a) = \lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h}$
- the **right-hand derivative** of f at $x = a$ is $f'_+(a) = \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}$.

$$f(x) = \begin{cases} x^2, & x \leq 1 \\ 2x-1, & x > 1 \end{cases} \quad \begin{matrix} \text{LHD} \\ \text{RHD} \end{matrix}$$

$$f'(x) = \begin{cases} 2x, & x \leq 1 \\ 2, & x > 1 \end{cases}$$

$$\therefore f'_-(1) = 2(1) = 2$$

$$f'_+(1) = 2$$

$$\therefore f'_-(1) = f'_+(1) = 2$$

Hence $f(x)$ is differentiable at $x = 1$

Therefore, $f(x)$ is continuous and differentiable at $x = 1$

Note: A function may be continuous at a given point but not necessarily differentiable at that point.



Question(2): Test for the continuity and differentiability of the following function at $x = 2$.

$$f(x) = |x - 2| = \begin{cases} x - 2, & x > 2 \\ -x + 2, & x \leq 2 \end{cases}$$

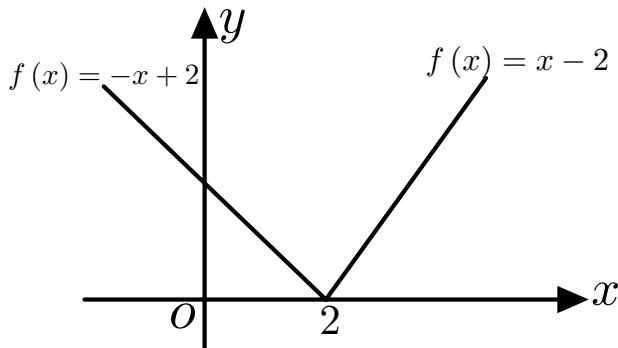
Stage (i): Continuity

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (-x + 2) = -2 + 2 = 0$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (x - 2) = 2 - 2 = 0$$

$$\text{Since } \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = 0$$

$\therefore f(x)$ is continuous at $x = 2$





Stage (ii): Differentiability

$$f(x) = |x - 2| = \begin{cases} x - 2, & x > 2 \\ -x + 2, & x \leq 2 \end{cases}$$

$$f'(x) = \begin{cases} 1 & , x > 2 \\ -1 & , x \leq 2 \end{cases}$$

$$\therefore f'_+(2) = 1 \quad \text{RHD}$$

$$f'_-(2) = -1 \quad \text{LHD}$$

$$\therefore f'_+(2) \neq f'_-(2)$$

Hence $f(x)$ is not differentiable at $x = 2$.

So, $f(x)$ is continuous but not differentiable at $x = 2$.

Practice Questions:

Q(1): Let $f(x) = \begin{cases} kx^2 - 1 & x < 2 \\ 3x + k & x \geq 2 \end{cases}$. Find the value of k for which $f(x)$ is continuous at $x = 2$.

A. $k = \frac{7}{3}$

B. $k = 7$

C. $k = \frac{3}{7}$

D. $k = 2$

Q(2): Consider the function $f(x) = \begin{cases} ax^2 + b & x \leq 1 \\ 2x - 1 & x > 1 \end{cases}$. For $f(x)$ to be differentiable at $x = 1$, what must be the values of a and b ?

A. $a = 2, b = -1$

B. $a = 1, b = 1$

C. $a = 0, b = 1$

D. $a = 1, b = 0$

Q(3): At which of the following values of x does the function $f(x) = |x^2 - 4|$ fail to be differentiable?

A. $x = 0$ only

B. $x = 2$ and $x = -2$ only

C. $x = 2, x = -2$, and $x = 0$

D. The function is differentiable for all real numbers x .





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End of Lesson

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